```
!pip install -q google-generativeai
import google.generativeai as genai
```

```
genai.configure(api_key="AIzaSyDvmXzyjXOHipKFddCzIUXpCWBFIb68fH0")
```

```
shared_memory = {
    "step_counter": 0,
    "max_steps": 5,
    "lock": None
}
```

```
def acquire_lock(agent_name):
    if shared_memory["lock"] is None:
        shared_memory["lock"] = agent_name
        return True
    return False

def release_lock(agent_name):
    if shared_memory["lock"] == agent_name:
        shared_memory["lock"] = None
```

```
def agent(agent_name):
    if not acquire_lock(agent_name):
        print(f"🔒 {agent_name} could not acquire lock. Skipping this cycle.")
        return

    current_step = shared_memory["step_counter"]
    if current_step >= shared_memory["max_steps"]:
        print(f"✅ {agent_name}: All steps completed.")
        release_lock(agent_name)
        return

    model = genai.GenerativeModel("gemini-2.5-flash")
    prompt = f"{agent_name} is executing step {current_step}. Describe what happens at this step."
    response = model.generate_content(prompt)

    print(f"\n🧠 {agent_name} executed step {current_step}:\n➡️ {response.text}")

    shared_memory["step_counter"] += 1
    release_lock(agent_name)
```

```
for i in range(10):
    print(f"\n--- Cycle {i+1} ---")

    # Reset lock before each cycle to simulate concurrency
    shared_memory["lock"] = None

    agent("Summarizer")
    agent("Analyst")
```

```
--- Cycle 1 ---

🧠 Summarizer executed step 0:
➡️ In the context of a "Summarizer," Step 0 typically represents the **initialization and setup phase**. This is where the summari

Here's a breakdown of what usually happens at this step:

1.  **Input Acquisition:**
    *   The most crucial action is **acquiring the source text** that needs to be summarized. This could involve:
        *   Reading from a file (e.g., `.txt`, `.pdf`, `.docx`).
        *   Accepting text directly as a string input (e.g., from an API or user interface).
        *   Scraping content from a URL.

2.  **Parameter and Configuration Definition:**
    *   The summarizer typically receives or retrieves instructions on *how* to summarize. This includes:
        *   **Desired summary length:** (e.g., "summarize to 3 sentences," "20% of original length," "maximum 100 words").
        *   **Summarization type:** (e.g., extractive vs. abstractive, if the system supports both).
        *   **Language:** (if the summarizer uses language-specific models or processing).
        *   **Output format:** (e.g., plain text, bullet points, JSON).
        *   Any other specific constraints or preferences.

3.  **Initial Pre-processing / Cleaning (High-Level):**
    *   While more in-depth NLP pre-processing might happen in subsequent steps, Step 0 might involve basic cleaning of the raw inp
        *   **Removing irrelevant characters:** (e.g., special symbols, excessive whitespace).
        *   **Stripping HTML tags:** If the input came from a web source.
        *   **Handling encoding issues:** Ensuring text is uniformly encoded (e.g., UTF-8).
        *   **Basic normalization:** (e.g., converting all text to a consistent case, though full tokenization often comes later).

4.  **Resource/Model Loading (for sophisticated summarizers):**
    *   For summarizers that use machine learning or complex NLP techniques, Step 0 might involve loading necessary resources:
        *   **Pre-trained language models:** (e.g., for abstractive summarization).
        *   **NLP libraries:** (e.g., spaCy, NLTK) if not already loaded globally.
        *   **Lexicons or dictionaries:** (for specific tasks like sentiment or keyword identification).

**In essence, Step 0 is about gathering all the raw materials, understanding the task requirements, and performing initial housekee

🧠 Analyst executed step 1:
➡️ When an analyst executes "Step 1," it's generally a **foundational and preparatory phase** that sets the stage for all subseque

Here's a breakdown of what typically happens:

1.  **Understanding the Objective & Scope:**
    *   **Receiving the Request/Problem Statement:** The analyst reviews the initial brief, request, project charter, or problem de
    *   **Clarifying the "Why" and "What":** They seek to understand the core purpose of the task. What problem are we trying to so
    *   **Defining Initial Scope:** They identify the boundaries of the task. What's included? What's explicitly out of scope (for
    *   **Asking Clarifying Questions:** This is crucial. The analyst will likely engage with stakeholders (manager, client, subjec

2.  **Initial Information Gathering & Research:**
```

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```
    *   **Accessing Initial Data Sources/Systems (if applicable):** For data-centric roles, this might mean identifying the relevan
```

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```
3.  **Tool & Environment Setup/Verification:**
```

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```
    *   **Verifying Readiness:** A quick check to ensure all initial prerequisites are met before diving into the core work.
```

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```
    *   **Formulating an Initial Approach:** Based on the objective and available information, the analyst starts to mentally (or w
```

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```
**In essence, Step 1 is about the analyst getting their bearings, ensuring they understand the mission, gathering their initial too

--- Cycle 2 ---

🧠 Summarizer executed step 2:
➡️ Assuming a typical multi-step process for an **extractive summarizer** (which selects important sentences directly from the sou

Here's a breakdown of what happens in this step:

1.  **Goal:** The primary goal of Step 2 is to identify and quantify various characteristics of each sentence (or sometimes words/p

2.  **Process & Common Techniques:**
    *   **Statistical Features:**
        *   **Term Frequency-Inverse Document Frequency (TF-IDF):** Calculating how unique and frequent words are in a sentence rel
        *   **Sentence Position:** Sentences at the beginning or end of paragraphs often carry topic-level information.
        *   **Sentence Length:** Longer sentences may contain more information, though very long ones can be verbose.
        *   **Numeric Data/Named Entities:** Presence of numbers, dates, people, organizations can signal factual importance.
        *   **Keyword Frequency:** Simple count of highly frequent words within a sentence.
    *   **Linguistic Features:**
        *   **Cue Phrases:** Identifying phrases like "In conclusion," "Therefore," "The main point is," "This study shows" that of
        *   **Lexical Chains/Semantic Cohesion:** Analyzing how words in different sentences are semantically related, helping to i
        *   **Part-of-Speech Tagging:** Identifying nouns, verbs, adjectives to focus on content words.
    *   **Graph-based Features:**
```